

TANU CHAUDHARY

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience in deep learning, computer vision, and Generative AI applications. Skilled in Python, PyTorch, TensorFlow, and Scikit-learn, with expertise in end-to-end ML pipelines, model deployment, REST APIs and production-ready AI solutions. Strong foundation in machine learning, data structures, algorithms and cloud technologies with a passion for building scalable and impactful AI systems.

EDUCATION

Vellore Institute of Technology, Chennai Master of Computer Applications CGPA: 8.62	Expected 2027
Galgotias University, Noida Bachelor of Computer Applications IO AI/ML CGPA: 8.6	2022 – 2025

EXPERIENCE

Google Generative AI Virtual Intern — AICTE	Jul 2024 – Sep 2024
- Designed AI automation pipelines using prompt engineering, reducing manual query handling by 30%. - Built foundational understanding of LLM inference pipelines and production-grade AI system architecture through hands-on project work.	

PROJECTS

Human Suspicious Activity Recognition System PyTorch · SigLIP 2 · Mamba · OpenCV GitHub
- Built a spatial-temporal deep learning pipeline using SigLIP 2 vision encoder and Mamba state-space model for multi-frame activity recognition in real-world video data. - Achieved 97% classification accuracy on real-world surveillance datasets and applied data preprocessing, augmentation, and Transformer-based feature extraction for robust inference.
Crime Prediction & Analytics Engine Python, XGBoost, Random Forest, FastAPI GitHub
- Engineered an end-to-end ML pipeline over 50,000+ crime records using ensemble models (XGBoost, Random Forest, Decision Tree), achieving 89% F1-score for hotspot prediction. - Built a FastAPI prediction pipeline with optimized preprocessing, reducing processing time by 25%.
AI Desktop Assistant Python, OpenCV, Gemini API, Speech Recognition, REST APIs GitHub
- Built a multi-modal AI agent combining face authentication (OpenCV), LLM-based response generation (Gemini API), and TTS automating 20+ OS-level tasks with context-aware interactions. - Designed concurrent architecture with REST API integration for real time voice and vision processing.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, PHP, SQL
ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transformers, GANs, Autoencoders
Generative AI & NLP: LLMs, Prompt Engineering, Hugging Face, NLP, Text Classification, Embeddings, AI in Applications
Computer Vision: OpenCV, Image Classification, Object Detection, Video Analysis, Feature Extraction
Data & MLOps: Pandas, NumPy, Matplotlib, Feature Engineering, Data Preprocessing, Model Evaluation, Docker, AWS (EC2, S3)
Networking & OS: TCP/IP Concepts, Linux/Unix, Operating System Concepts, Virtualization Basics, Cloud Computing (AWS)
Databases: MySQL, MongoDB, Oracle (basics), SQL, NoSQL, Database Management
Backend & Tools: FastAPI, Flask, REST APIs, Git, GitHub, Jupyter, Google Colab, VS Code, DSA, OOP, System Design

CERTIFICATIONS

AI & ML Engineering — Microsoft | [Link](#)
Machine Learning Foundations — AWS | [Link](#)
Machine Learning & Data Science — GeeksforGeeks | [Link](#)
Oracle Fusion AI Agent Studio — Oracle | [Link](#)
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle | [Link](#)

ACHIEVEMENTS

- Research: Quantum-Inspired Multi-Agent Learning for Blue Carbon Modeling — Under Review, Springer.
- Smart India Hackathon Pre-Qualifier — 2023 and 2024.